Relational SQL Queries

The information you want isn't always present in ONE TABLE alone, so *you have to* **go to** another table.

This is done via the existence of **Keys.**

- There are **2 main types of keys**, *Primary & Foreign*

Primary is a UNIQUE identifier in a table, **for example**

- The producer table has **producer → producerID (**Primary Key**)**

But then, you can see that prodcuerID is ALSO in the *Album table*?

- Then it's the **F**OREIGN **K**EY, these two are paired up to *match* ROWS of data *across* **DIFFERENT** *tables*.

That may sound confusing, but **let's do an example,** it should *clear it* up.

- Let's grab the *Producer's Name* for the Album **Starlight**

__Notice how we have to say table.column__, since we're working with keys there might be duplicates in __different tables__, so you need to mention which table.

```sql
SELECT Producer.producername -- Columns I want (I want producer name)

FROM Producer -- One of the two tables you wish to match (doesn't matter which)

INNER JOIN Album -- "INNER JOIN" combines tables by the key combonation (producerID)

   ON Producer.producerID = Album.producerID -- Match our keys

WHERE Album.albumtitle = 'Starlight' -- Filter, for anything in either table.
```

With RELATIONAL QUERIES were now able to **GRAB** the *RELATED* producer for our album, **DESPITE** *it not being in the same table.*

That may STILL look confusing, Let me show you what the **INNER JOIN** *actually does.*

Let's print *EVERY column* from our INNER JOIN

- *(Replace the **first line** with SELECT *)*

	producerid	producername	studio	yearstarted	albumid	albumtitle	releaseyear	genre	tracks	sales	producerid
1	4	Max Martin	Maratone	1995	102	Starlight	2022	Pop	15	3200000	4

You can see, it COMBINED both tables to make a *new table* with all **columns in one.**

- So it just becomes like a **normal filter like in part 2&3** *of the lab*.

Actually, we can join MORE than one table via the keys.

- *Let's find the fee for the* ***STARLIGHT*** *Album,* fee exists in **PERFORMANCES**, *so we need to be able to* **INNER JOIN** *that table too.*
- Also, SELECT the fee column from PERFORMANCES, **since thats what we want**

Take the *previous query,* and add this before the **WHERE line**

```
INNER JOIN PERFORMANCES
    ON PERFORMANCES.albumid = Album.albumid
```

```
SELECT PERFORMANCES.fee -- Columns I want (I want producer name)
FROM Producer -- One of the two tables you wish to match (doesn't matter which)
INNER JOIN Album -- "INNER JOIN" combines tables by the key combonation (producerID)
    ON Producer.producerID = Album.producerID -- Match our keys
INNER JOIN PERFORMANCES -- Add the new TABLE
    ON PERFORMANCES.albumid = Album.albumid -- Match the other key
WHERE Album.albumtitle = 'Starlight' -- Filter, for anything in either table.
```

What we're doing here, is finding ANOTHER **Primary Key & Foreign Key** combo

- (albumid)

Here we can see the fully joined table, from which we **filter by the name,** and select to **only get the fees.**

	producerid	producername	studio	yearstarted	albumid	albumtitle	releaseyear	genre	tracks	sales	producerid	artistid	albumid	fee
1	4	Max Martin	Maratone	1995	102	Starlight	2022	Pop	15	3200000	4	101	102	150000

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*That's actually really it for BASIC relational SQL queries, next part will cover **more complex filters**.*